Lab 3: Physical Coding Sublayer (PCS) 64b/66b Encoding & Scrambling

Hardware-in-the-Loop Simulation, Self-Synchronizing LFSR & 66-Bit Waveform Analysis

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1. System Architecture & 64b/66b PCS Datapath

Hardware-in-the-Loop Setup:

- **Host Interface**: `tap0` (`10.0.0.1/24`, MAC `02:00:00:00:00:01`).
- **Target Interface**: `tap1` in namespace `ns_b` (`10.0.0.2/24`, MAC `02:00:00:00:00:02`).
- **RTL Stack (`top.sv`)**: MAC TX/RX, 64b/66b PCS encoder/decoder, and 58-bit parallel LFSR scrambler.
- **C++ Driver (`wrapper.cpp`)**: Non-blocking TAP driver with synchronous clocking across dual Linux namespaces.
- **Specification**: IEEE 802.3 Clause 82 (40G/100GBASE-R).

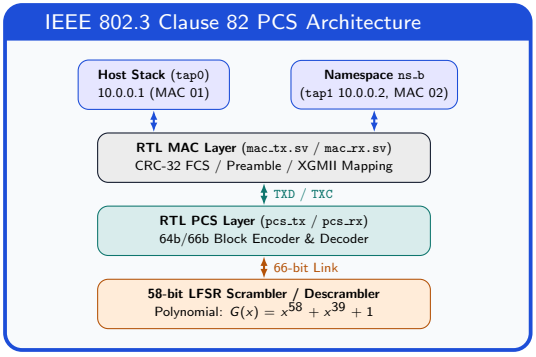

Figure 1: *Hardware pipeline from Linux TAP devices to the 66-bit physical link.*

2. Environment Setup & Traffic Generation

Execution Sequence:

1. **Configure TAP Interfaces**:
   `sudo ./setup_tap.sh`
   Configures tap0 and namespace ns_b with tap1.
2. **Build and Run Verilator Emulator**:
   `verilator --cc --trace --exe --build wrapper.cpp top.sv`
   `sudo ./obj_dir/emulator --enable-scrambler`
3. **Capture Traffic**:
   `sudo tcpdump -i tap0 -w lab3_capture.pcap`
4. **Send Patterned Ping**:
   `ping -c 2 -p cafe0003 -I tap0 10.0.0.2`

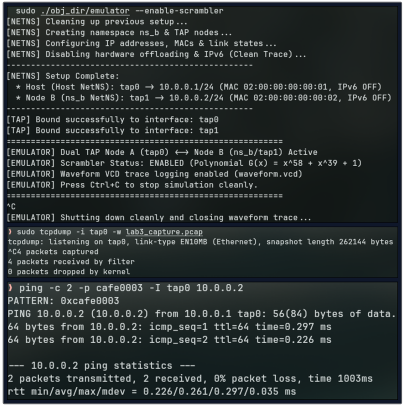

Figure 2: *Terminal execution showing TAP binding, Verilator build, and no packet loss.*

3. Layer 2 / Layer 3 Wireshark Packet Analysis

Captured ICMP Echo Request Frame (98 Bytes Total):

- **Layer 2 Header (14B)**: Dst MAC `02:00:00:00:00:02`, Src MAC `02:00:00:00:00:01`, EtherType `0x0800` (IPv4).
- **Layer 3 Header (28B)**: IPv4 `10.0.0.1` → `10.0.0.2` (20B, Len 84B, ID `0x7F6D`), ICMP Type 8 (Echo Request, Code 0, Checksum `0xDB80`, ID `0x7602`, Seq 1).
- **Payload (56B)**: 16-byte kernel timestamp + 40-byte repeating pattern `ca fe 00 03`.

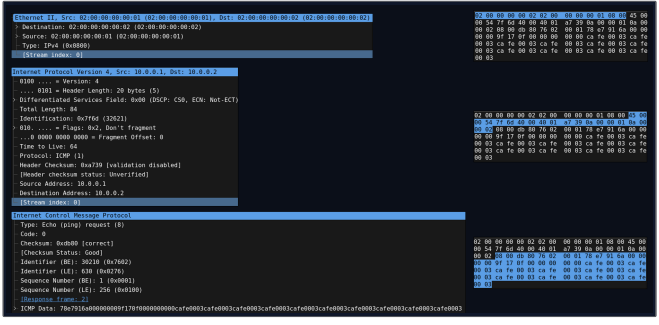

Figure 3: *Wireshark packet dissection and correlated hex dumps.*

4. Inter-Packet Gap (IPG) & Control Block Encoding (BTF = 0x1E)

Idle Control Block Compression:

- **Idle Input**: MAC outputs TXD = 0x0707070707070707 with TXC = 8'hFF, meaning all 8 lanes have an control signal.
- **Control Sync Header**: PCS encoder sets Sync = 2'b10 (raw_hdr = 2'b10) for Control Blocks.
- **Payload Compression**: Sets BTF = 0x1E (bits [7:0]) and compresses 8 idle bytes into 7-bit PCS codes (raw_payload = 64'h0000000000000001E).

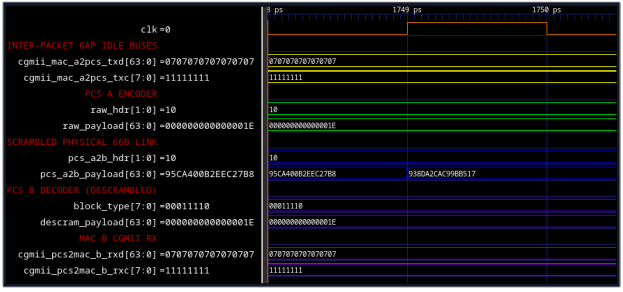

Figure 4: *Idle compression: 8 idle bytes encoded into Control Block (BTF = 0x1E, Sync = 2'b10).*

5. Frame Start Delimiter (/S/) & Control Block Framing (BTF = 0x78)

Frame Start Delimiter & Block Translation (Cycle 1–2):

- **Start Delimiter Detection**: MAC asserts /S/ = 0xFB on Lane 0 with TXC = 8'h01 (TXD = 0xD5555555555555FB).
- **Control Block Type 0x78**: PCS encoder outputs Sync = 2'b10 and sets BTF = 0x78 on Byte 0.
- **Preamble Passing**: Preamble (0x55) and SFD (0xD5) pass into payload bits [63:8] (raw_payload = 0xD555555555555578).

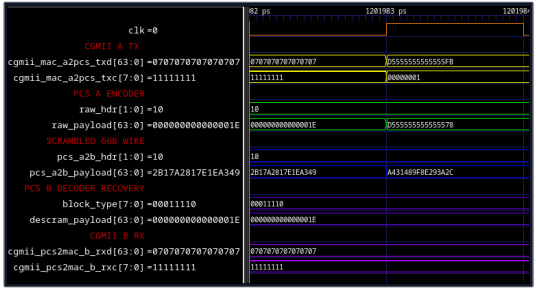

Figure 5: *Transition from Idle to Frame Start (/S/ = 0xFB) on Lane 0 mapped to BTF = 0x78.*

6. SW-to-HW Correlation: Payload Data Blocks (`Sync = 2'b01`)

64-Bit Data Streaming & Header Decoding (Cycles 3–12):

- **Data Block Framing**: For data words (`TXC = 8'h00`), PCS encoder sets `Sync = 2'b01` without BTF overhead.
- **Direct Word Mapping**: Unscrambled payload matches Wireshark byte-for-byte (L2 MACs, IPv4 ID `0x7F6D`, ICMP Checksum `0xDB80`).
- **Throughput**: Transmits 8 bytes/cycle at 100% line rate with 3.125% framing overhead.

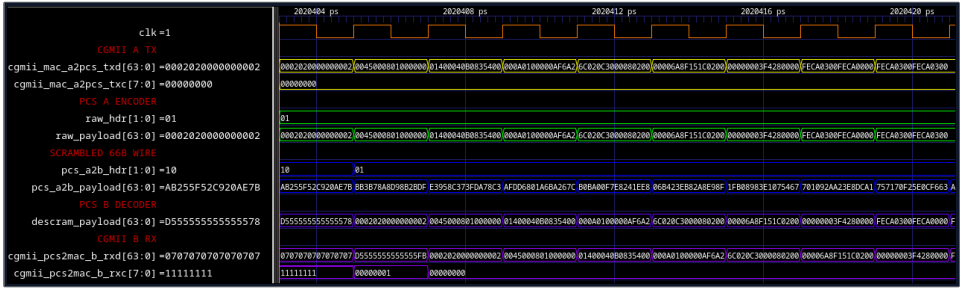

Figure 6: *Cycle-by-cycle alignment of L2/L3 packet headers across 64-bit Data Blocks (`Sync = 2'b01`).*

7. 58-Bit LFSR Scrambler & Sync Header Invariance

Self-Synchronizing Scrambler Pipeline ($G(x) = x^{58} + x^{39} + 1$):

- **Bit Recurrence**: $y[i] = x[i] \oplus y[i-39] \oplus y[i-58]$, computed in 1 cycle via self-synchronizing LFSR.
- **Sync Header Invariance**: Sync header `Sync[1:0]` bypasses the LFSR to preserve receiver block alignment.
- **Descrambler Recovery**: Descrambler recovers payload without sync tokens: $x'[i] = y[i] \oplus y[i-39] \oplus y[i-58] \equiv x[i]$.

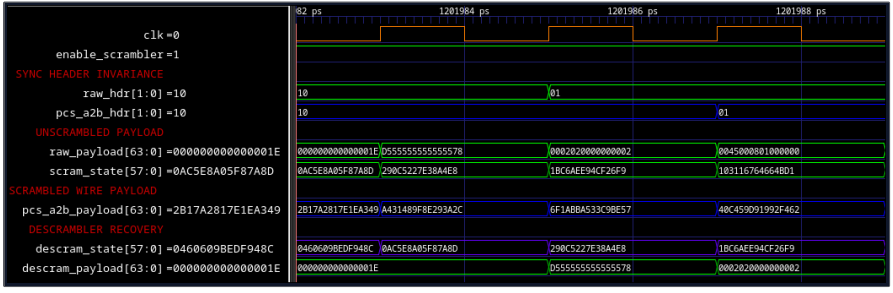

Figure 7: *Payload bit randomization via $G(x) = x^{58} + x^{39} + 1$ and exact descrambler recovery.*

8. Mathematical Proof of Self-Synchronization

Definitions (IEEE 802.3 Clause 49/82):

- **Transmitter (y)**: Encodes data bit $x[i]$ using past wire bits:

$$y[i] = x[i] \oplus y[i-39] \oplus y[i-58]$$

- **Receiver (x')**: Decodes incoming wire bit $y[i]$ using the same taps:

$$x'[i] = y[i] \oplus y[i-39] \oplus y[i-58]$$

Algebraic Proof:

$$x'[i] = \underbrace{\big(x[i] \oplus y[i-39] \oplus y[i-58]\big)}_{\text{Substitute } y[i]} \oplus y[i-39] \oplus y[i-58]$$

$$= x[i] \oplus \big(y[i-39] \oplus y[i-39]\big) \oplus \big(y[i-58] \oplus y[i-58]\big) \quad \text{(Associative)}$$

$$= x[i] \oplus 0 \oplus 0 \quad (\text{Since } A \oplus A = 0)$$

$$\equiv \mathbf{x[i]}$$

Takeaway: The receiver recovers clean data with 0 bit errors after 58 bits of wire traffic.

8. Frame Termination (/T/) & BTF Selection (0xE1)

Frame Termination Delimiter & CRC-32 FCS (Cycle 16):

- **CRC-32 FCS**: MAC computes CRC-32 ($P(x) = x^{32} + x^{26} + \cdots + 1$) over 98 bytes → 0x45210EBC (Lanes 2–5: BC 0E 21 45).
- **Clause 82 BTF Selection**: Delimiter /T/ = 0xFD on Lane 6 (TXC = 8'hC0) maps to BTF = 0xE1 (raw_payload = 0x0045210EBC0300E1).
- **Hardware Reconstruction**: Decoder detects BTF = 0xE1, restores TXC = 8'hC0, and MAC RX validates the FCS.

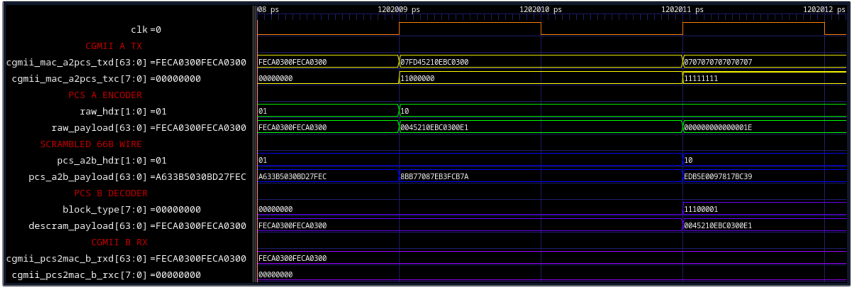

Figure 8: *Cycle 16 frame termination: /T/ on Lane 6 mapped to BTF = 0xE1 with CRC-32 FCS (0x45210EBC).*

Thank You!